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# Psymple: A Python package for complex systems modelling

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## Summary

The modelling of complex systems, which are characterised by high degrees of interdependence amongst components, is continually evolving in response to developing computing and visualisation power. Examples such as ecological, economic or social systems are classically modelled using statistical or correlative techniques, which have natural shortcomings when used for predictive modelling outside of the range of their parameterisation data.

Predictive modelling is improved by deterministic models which capture the evolution of a system, or by hybrid models composed of both deterministic and correlative parts. These models often feature significantly increased size and complexity, and there has been a growing requirement for so-called next generation modelling platforms which are built to a specification requiring modular and data-first implementations which drive clear, adaptable, reusable and accessible modelling practices.

The platform `psymple` is designed to facilitate the development of hybrid complex system models and modelling frameworks. It allows users to link together arbitrary combinations of modular differential equation systems and functional components to build categorical diagrams representing a complex system. A compilation process automatically generates simulatable system equations from these diagrams using the symbolic mathematics package `sympy`. Ultimately, this allows users to focus on the components and interactions of models, rather than their complex equation structure.

## Statement of need

The development of `psymple` emerged from the complex system modelling requirements of ecological systems. Ecological niche models, also called species distribution models (SDMs) predict species distributions in response to environmental variables such as geographic and climatic features (Elith & Franklin, 2017; Elith & Leathwick, 2009). Classically, these models are formed using correlative or statistical approaches which match observational data to a set of environmental variables to produce favourability ranges for each species.

An alternative approach is mechanistic modelling, which instead of observational data use physiological data to capture the underlying mechanisms which drive species distribution, such as energy balance, population dynamics or response to climate (Kearney & Porter, 2009). In contrast to correlative approaches, mechanistic SDMs decouple the physiology of a species from their native geography or climate, and allow SDMs in new geographic or climatic regimes to be created in the absence of observational data (Johnston et al., 2019).

An example mechanistic framework is physiologically-based demographic modelling (PBDM), which creates holistic ecosystem models based on the weather-driven biology of component species, allowing for predictive phenology, age- or mass-structured population dynamics, and distribution assessments, see (Gutierrez & Ponti, 2022a, 2022b) for an overview and further references. With this approach, PBDM can account for tritrophic ecosystem interactions (Gutierrez et al., 1999), or model the effects of climate change (Gutierrez et al., 2023).

The schools of thought around correlative SDMs and mechanistic frameworks such as PBDM are seen as largely disjoint (Dormann et al., 2012), but their development can be traced back to early common roots (Fitzpatrick & Nix, 1970; Gutierrez et al., 1974; Wit & Goudrian, 1978). A component of the PBDM framework is the use of physiological data to parametrise "biodemographic" functions capturing biophysical or biochemical mechanisms, such as the development, mortality and fecundity rates of a species in response to environmental variables (Ponti & Gutierrez, 2023).

The use of biodemographic functions in PBDM combines the considerable holistic advantages of mechanistic SDMs, while retaining the comparable ease of parametrisation as status-quo correlative models. More widely, there has been growing interest and development of ecological models explicitly composed of both correlative and mechanistic components (Buckley et al., 2010; Tourinho & Vale, 2023), combining the benefits of both areas. In the wider context of complex systems modelling, the approach of building composite models out of different techniques is called hybrid, or spectrum, modelling.

While PBDM is a highly-developed framework, it shares the same barriers to widespread introduction as general hybrid complex system models. These barriers include the lack of available modelling frameworks (Buckley et al., 2018), the lack of flexibility in existing models (Buckley et al., 2010), or the lack of modelling platforms to implement existing ideas (Ponti & Gutierrez, 2023). The package `psymple` is a general platform designed to facilitate the creation of hybrid complex systems models and modelling frameworks.

Models in `psymple` are built from arbitrary combinations of modular mechanistic, dynamic components and correlative, functional components which naturally interact with each other. This structure allows for the systematic implementation of modelling frameworks such as PBDM, and, more widely, those capturing biological, economic and social systems, for which it is not feasible to capture the laws of interaction purely mechanistically. Examples include agroecological, bioeconomic and Earth systems modelling, and the development of `psymple` is a necessary first step in the development and release of accessible and impactful tools in these areas.

# Description

The workings of `psymple` are based on the ideas of the category-theoretic dynamic systems modelling package `AlgebraicJulia.jl` (Libkind et al., 2024), in which modular 'ported' objects, called *resource sharers*, containing ordinary differential equations, expose internal variable states at different ports. Ports are linked by wires which represent the sharing of state variables across the dynamics equations of the system to create complex interacting dynamical systems. Mathematically, resource sharers form algebras over the operad of undirected wiring diagrams, see (Libkind et al., 2022).

In `psymple`, these ideas are extended to realise functions as ported objects, called functional ported objects (FPOs), alongside resource sharers, which in `psymple` are called variable ported objects (VPOs). FPOs contain systems of multivariate functions, whose values are exposed at output ports, and whose arguments can read information in from other FPO outputs or from VPO variable ports. The structure of input ports is also introduced to VPOs by allowing functions defining differential equations to read information in the same way as FPOs.

FPOs and VPOs can be arbitrarily combined into composite ported objects (CPOs) through

the concepts of directed wires, which capture partial substitution of functional information, and variable wires, which implement the functionality of resource sharing. CPOs can themselves read and expose information from input, output and variable ports to create fully modular and arbitrarily complex hybrid systems of both functional and dynamic components whose nested hierarchy reflects the model structure.

More generally, `psymple` is built to a specification shared by "next-generation" dynamical systems modelling frameworks, see (Baez et al., 2023), including being *faceted*, where models can be considered one piece at a time; *modular*, where components naturally compose together; and *functorial*, where the data describing the model (its syntax) is systematically and reliably transformed into system behaviour (its semantics). These ideas allow for legible modelling of highly complex, specialised systems, and drive clear, adaptable and accessible modelling practices in line with those promoted by the Open Modeling Foundation. For example, modular structures pave the way for cross-platform integrations and promote reuse and flexibility, while an abstracted data structure enables the creation of low- or no-code interfaces to improve utilisation amongst non-specialist users.

Concretely, the faceted, modular structure provided by ported objects in `psymple` is complemented by a string-based authoring interface which codifies the diagrammatic wiring structures, and enables no-code authoring from Python dictionary objects or `JSON` file formats. The data-based syntax defining ported objects is internally processed into model semantics, meaning simulatable systems of equations, through a composition process using the Python symbolic mathematics package `sympy` (Meurer et al., 2017). This enables the automatic collection and simplification of equations, the elimination of errors from manually combining complex system equations, and clear inspection of a whole system, or its parts, with automatic outputs including LaTeX format.

# Research

The development of `psymple` emerged as part of a collaborative effort between IDEMS International and CASAS Global to increase the uptake, accessibility and impact of the physiologically-based demographic modelling (PBDM) framework. The development of `psymple` is a necessary first step to capture PBDM in sufficient generality to enable modular and flexible implementations of existing and new models into the framework to drive future research and impact potential.

The specification of `psymple` was created from the "collaborative modelling" approach headed-up by the Topos Institute, which uses the mathematical field of applied category theory to capture the mechanisms required to create modelling frameworks collaboratively, accounting for diversity in technical language, data availability and subject expertise. This specification, coupled with the generality and flexibility of `psymple`, enables frameworks beyond PBDM to be developed or implemented, and pushed towards new research and impacts.

For example, research is currently active in bioeconomic modelling, which extends the supply-demand and mass-structured mechanisms of PBDM to understand the economic interaction that humans have with agroecological systems, see (Gutierrez et al., 2020; Regev et al., 1998). More widely, ideas are being conceived around applications in Earth systems modelling, multi-layer modelling incorporating feedback loops to modelling techniques such as collaborative agent-based modelling, and data management and source control across collaborative practices.

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